

Flow Dynamics of a Converging-Diverging Nozzle Under Varying Nozzle Pressure Ratio

Project Summary — CFD Research (ANSYS Fluent)

Based on unpublished research conducted with a student research team, Department of Aeronautical Engineering, GIT Belagavi. This is a summary of the project, not the manuscript.

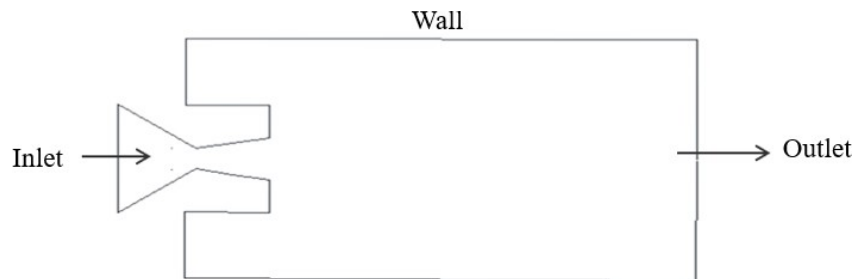
Problem & Motivation

Converging-diverging (CD) nozzles are used everywhere high-speed flow needs to be accelerated past the speed of sound: rocket engines, supersonic wind tunnels, and high-speed aircraft. How well a CD nozzle performs depends heavily on the Nozzle Pressure Ratio (NPR), the ratio of the reservoir pressure feeding the nozzle to the ambient pressure it exhausts into. Get the NPR wrong relative to the nozzle's geometry and the flow either over-expands or under-expands, both of which generate shock waves, pressure losses, and noise instead of clean thrust.

This project set out to characterize how a specific CD nozzle geometry behaves as NPR is varied, with a focus on how the shock structure and Mach number distribution change across the nozzle length. The goal was to build a clear, simulation-based picture of where and how this nozzle transitions between under-expanded, correctly expanded, and over-expanded flow.

Methodology

The nozzle was modeled as a 2D domain in ANSYS Fluent, split into inlet, throat, outlet, and exhaust regions, and discretized with a structured mesh that was refined in the regions where shock waves were expected, to keep those features well resolved. The simulation used a density-based solver with second-order upwind discretization, solving the compressible flow, energy, and k-epsilon turbulence equations to capture shock formation and pressure variation across the nozzle.

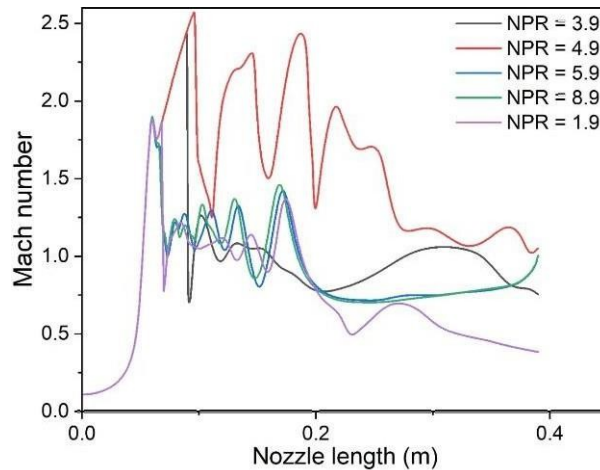


2D schematic of the CD nozzle domain used for simulation (inlet → throat → outlet → exhaust).

Simulations were run across a range of NPR values, from under-expanded to over-expanded conditions, to compare how the flow field, shock structure, and pressure decay changed with each case.

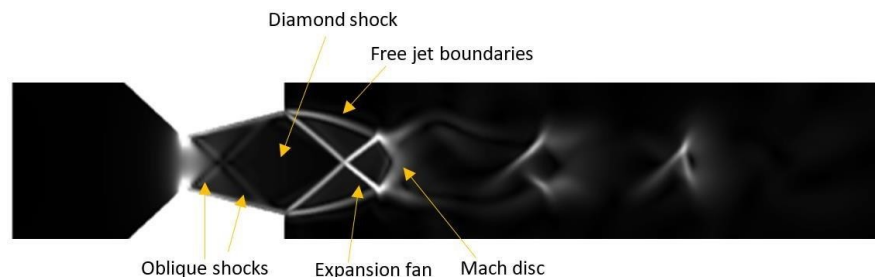
Key Results

The Mach number profile along the nozzle length showed a clear, repeating rise-and-fall pattern at higher NPR values, evidence of the shock-cell structure that forms when exit pressure and ambient pressure don't match. As NPR increased, the profile showed sharper Mach number spikes and a longer, more oscillatory shock-cell pattern before the flow settled, while lower NPR cases showed a smoother, faster decay.



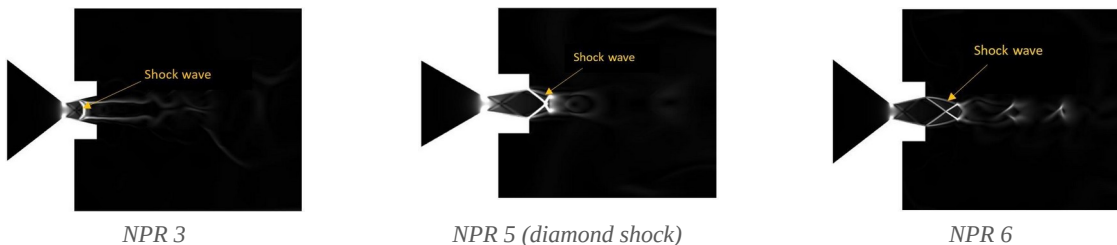
Mach number variation along the nozzle length across five NPR conditions (1.9 to 8.9).

Numerical schlieren visualization made the shock structure itself visible: oblique shocks forming near the nozzle exit, expansion fans, and, at higher NPR, the shocks merging into a Mach disk at the center of the jet, flanked by the diamond-shaped shock pattern characteristic of an under-expanded jet.



Numerical schlieren image showing oblique shocks, expansion fan, Mach disc, and free jet boundaries.

Comparing individual NPR cases side by side showed how sensitive the shock pattern is to pressure ratio: the number and spacing of visible shock cells changed noticeably between NPR 3, 5, and 6, with the classic diamond-shock pattern appearing most clearly around NPR 5 for this nozzle geometry.



What This Demonstrates

- Applied CFD skills: setting up a compressible-flow simulation in ANSYS Fluent, including domain definition, structured meshing with targeted refinement, solver selection, and turbulence modeling.
- Aerodynamics fundamentals in practice: connecting a single parameter (NPR) to real, physically meaningful outcomes, shock structure, Mach number distribution, and flow stability, rather than treating the simulation as a black box.
- Data interpretation: reading simulation output (Mach number plots, schlieren visualization) to explain why the flow behaves differently across conditions, not just reporting that it does.
- Relevance beyond aerospace: the underlying skill set, translating a technical parameter sweep into a clear before/after comparison and a concise explanation of the result, is the same skill used in any engineering or program-management role that involves evaluating design trade-offs.